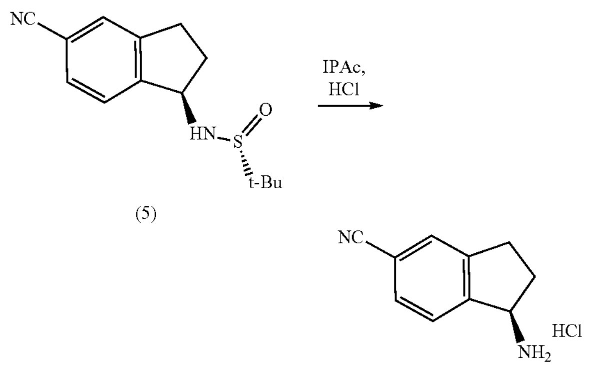
20-30° C. for 8-10 hours to give 42.2 kg (64.8%) of (R)—N—((R)-5-cyano-2,3-dihydro-1H-inden-1-yl)-2-methylpropane-2-sulfinamide.

Example 3: Synthesis of (R)-1-amino-2,3-dihydro-1H-indene-5-carbonitrile hydrochloride

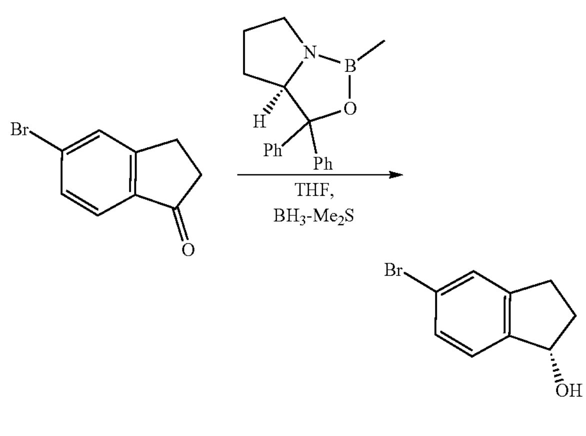


(5)

(4)

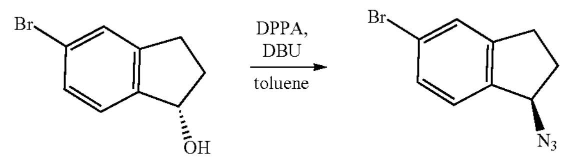
Isopropyl acetate (IPAc, 942 kg, 1082.8 L) was added to (R)—N—((R)-5-cyano-2,3-dihydro-1H-inden-1-yl)-2-methylpropane-2-sulfinamide (42.2 kg, 160 8 mol, 1.00 equiv.) and the solution was treated with 6M HCl (39 kg, 197.7 mol, 1.23 equiv.) for a period of 16-20 hours until IPC-3 was satisfied; residual (R)—N—((R)-5-cyano-2,3-dihydro-1H-inden-1-yl)-2-methylpropane-2-sulfinamide ≤0.5%. Crude (R)-1-amino-2,3-dihydro-1H-indene-5-carbonitrile hydrochloride was isolated by centrifuge. The wet cake was washed with IPAc (130 kg 149.4 L) and dried at 30-40° C. for 16-24 hours and the LOD was measured (report, result: 0.28% w/w). The (R)-1-amino-2,3-dihydro-1H-indene-5-carbonitrile hydrochloride product was analyzed for purity (specification: ≥98.0% area, result: 99.7% area), chiral purity (specification: ≥99.0%, result: 99.7% area) and Karl Fischer titration (report, result: 0.45% w/w). The title compound was isolated in 91.1% yield.

Example 4: Synthesis of (S)-5-bromo-2,3-dihydro-1H-inden-1-ol



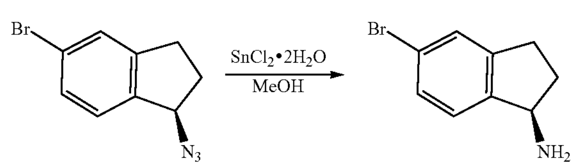
To a solution of 5-bromo-2,3-dihydro-1H-inden-1-one (50 g, 237 mmol, 1.0 equiv) in THF (400 mL) under a nitrogen atmosphere was added (3R)-1-methyl-3,3-diphenyl-hexahydropyrrolo[1,2-c][1,3,2]oxazaborole (37 mL of 1 M in toluene, 0.15 equiv). The mixture was cooled to −10° C. and borane dimethyl sulfide (10 M in THF) (32.2 g, 1.4 equiv) was added dropwise with stirring over 1 h. After stirring for 3 h at −10° C., the reaction was quenched by slow addition of water (200 mL). The resulting solution was extracted with ethyl acetate (200 mL) three times. The combined organic layers were washed with brine (300 mL), dried over anhydrous sodium sulfate, and concentrated. The residue was purified using a silica gel column packed with 1% TEA in petroleum ether (30% ethyl acetate/petroleum ether) to give a solid that was triturated with hexane (300 mL) to afford 38.0 g (75%) of (1S)-5-bromo-2,3-dihydro-1H-inden-1-ol as a light yellow solid. LRMS (ES): calculated for C_9H_9BrO, 212.0 Da, measured 195 m/z $[M+H-18]^+$.

Example 5: Synthesis of (R)-1-azido-5-bromo-2,3-dihydro-1H-indene



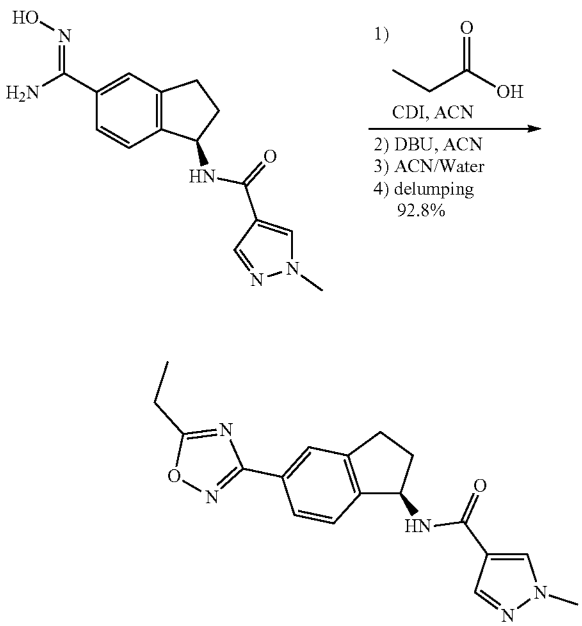
To a solution of (S)-5-bromo-2,3-dihydro-1H-inden-1-ol (42 g, 197 mmol, 1.0 equiv) in toluene (500 mL) was added diphenylphosphorylazide (74.3 g, 270.0 mmol, 1.4 equiv) under nitrogen. To this mixture was added DBU (45 g, 295 mmol, 1.5 equiv) dropwise with stirring at 0° C. over 1 h. After stirring for 3 h between 0 to 15° C., the mixture was diluted with ethyl acetate (400 mL) and washed with water (400 mL) three times. The organic layer was dried over anhydrous sodium sulfate, concentrated, and purified using a silica gel column packed with 1% TEA in petroleum ether (eluted with petroleum ether) to give 44.4 g (95%) of (R)-1-azido-5-bromo-2,3-dihydro-1H-indene as dark brown oil. The dark brown oil was used in next step without further purification. LRMS (ES): calculated for $C_9H_8BrN_3$, 237.0 Da, measured 195, 197 m/z $[M+H-42]^+$.

Example 6: Synthesis of (R)-5-bromo-2,3-dihydro-1H-inden-1-amine



To a solution of (R)-1-azido-5-bromo-2,3-dihydro-1H-indene (44.3 g, 186 mmol, 1.0 equiv) in methanol (600 mL) was slowly added $SnCl_2 \cdot 2H_2O$ (76 g, 337 mmol, 1.81 equiv). After stirring overnight at rt, the mixture was diluted with ethyl acetate (500 mL) and NaOH (2 N, 700 mL), stirred at rt for 1 h, and filtered. The filtrate was separated and the aqueous layer was extracted with ethyl acetate (300 mL). The combined organic layers were extracted with HCl (1 N, 500 mL) twice and the aqueous layers were combined.

Example 12: Synthesis of (R)—N-(5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide



A reactor (vessel 1) was charged with CDI (11.9 kg, 73.67 mol, 1.05 equiv.) and CH_3CN (131.4 kg, 167.2 L) and the resulting mixture was treated with propionic acid (5.7 kg, 77.18 mol, 1.10 equiv.) maintaining a temperature of <25° C. The transfer line was rinsed with CH_3CN (10.2 kg, 13.0 L) and the rinse was transferred the bulk reaction mixture. The resulting clear solution was stirred at 20±5° C. for at least 1 h. The reaction was considered to be complete when the IPC was met (specification: free propionic acid ≤20 mol % by ^1H NMR; result: 4.8% free propionic acid).

A separate reactor (vessel 2) was charged with (R)—N-(5-(N'-hydroxycarbamimidoyl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide (21.0 kg, 70.16 mol, 1.00 equiv.) and CH_3CN (40.4 kg, 51.4 L). The freshly prepared active imidazole solution in vessel 1 was transferred to vessel 2. Vessel 1 was rinsed with CH_3CN (20.2 kg, 25.7 L) and the rinse was transferred to added to vessel 2. The mixture was heated to 50±5° C. The reaction was stirred at this temperature for at least 12 hours (actual reaction time: 16.6 hours). The mixture was a slurry that as easily stirred. An IPC sample was taken (specification: (R)—N-(5-(N'-hydroxycarbamimidoyl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide ≤2%, result: 0.28%).

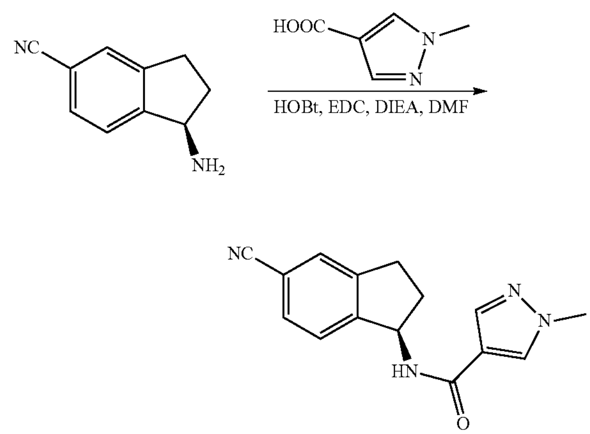
The reaction was then charged with DBU (21.4 kg, 140.32 mol, 2.00 equiv.). The transfer line was rinsed with CH_3CN (6.5 kg, 8.3 L) and the rinse was transferred to the bulk solution. The temperature was adjusted to 70±5° C. and the mixture as stirred at 70±5° C. for at least 2 h until the IPC was met (specification: imidazole intermediate ≤2 area %; result: 0.16% imidazole intermediate).

The reaction mixture was quenched with the addition of water (64.6 kg) while maintaining the temperature ≥50° C. The temperature was adjusted to 55±5° C. and polish filtered. The solution was concentrated at a temperature of ≤50° C. but not below 10° C. until the batch volume was ~200 L. The thick slurry was heated to 75±5° C. and the clear solution was diluted with water (417.3 kg) while maintaining the temperature ≥70° C. The temperature was adjusted to 75±5° C. and the mixture was stirred at 75±5° C. for 2 h. Then the mixture was then slowly cooled to 20±5° C. over a period of at least 4 h. The mixture was stirred 20±5° C. for at least 2 h (actual reaction time: 11.5 h).

The solid was collected by filtration and the wet cake was washed with water (3×161.5 kg). The solid was dried in a vacuum oven at ≤50° C. with a slow nitrogen bleed for at least 24 h (actual drying time: 48 h) and analyzed for LOD (specification: LOD ≤1% w/w; result: LOD 0.05% w/w).

The dried (R)—N-(5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide was analyzed for purity (specification: ≥97% area; result: 100% area). A total of 22.0 kg of (R)—N-(5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide (92.8% yield) was obtained. The product (22.0 kg) was then de-lumped to give 21.4 kg (97.3%).

Example 13: Synthesis of (R)—N-(5-cyano-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide



To a mixture of 1-methyl-1H-pyrazole-4-carboxylic acid (2.3 g, 18.2 mmol, 1.2 equiv), HOBt (2.1 g, 15.1 mmol, 1.0 equiv), and EDC (5.8 g, 30.3 mmol, 2.0 equiv) in DMF (10 mL) was added DIEA (7.5 mL, 45.4 mmol, 3.0 equiv). The mixture was stirred for 10 min, followed by addition of (R)-1-amino-2,3-dihydro-1H-indene-5-carbonitrile hydrochloride (2.9 g, 15.1 mmol, 1.0 equiv). The reaction mixture was stirred overnight and then diluted with water (60 mL). The solid was collected, washed with water (20 mL), and dried to give 3.5 g (86%) of (R)—N-(5-cyano-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide as an off-white solid. ^1H NMR (400 MHz, methylene chloride-d_2) δ 7.86 (s, 1H), 7.74 (d, J=0.8 Hz, 1H), 7.60-7.48 (m, 2H), 7.48-7.42 (m, 1H), 6.06 (d, J=8.4 Hz, 1H), 5.69 (q, J=8.3 Hz, 1H), 3.94 (s, 3H), 3.15-2.90 (m, 2H), 2.74-2.64 (m, 1H), 2.03-1.90 (m, 1H). LRMS (ES): calculated for $C_{15}H_{14}NO$, 266.1 Da, measured 267.1 m/z $[M+H]^+$.